WORKED SOLUTION

STEP III 2011, Question 13

Probability and Statistics • Mode of Discrete Distributions

STEP III 2011, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s3-q13

The key idea

To find the most likely value of a discrete random variable, we don't differentiate: we compare neighbouring probabilities. The ratio $\mathrm{P}(X = r+1)/\mathrm{P}(X = r)$ is a simple rational function of r, and asking when it exceeds 1 turns into a linear inequality in r. The probabilities rise while the ratio is above 1 and fall once it drops below, so the peak sits at the last r before the switch. The same method works with and without replacement; only the ratio changes.

Solution

Part (i)

With replacement, the draws are independent and each is black with probability $p = b/n$, so $X \sim \mathrm{B}(k, p)$:

$$\mathrm{P}(X = r) = \binom{k}{r} p^r (1-p)^{k-r}, \qquad r = 0, 1, \ldots, k.$$

Assume first that $0 < b < n$, so that every one of these probabilities is positive. For $0 \le r \le k-1$,

$$\frac{\mathrm{P}(X = r+1)}{\mathrm{P}(X = r)} = \frac{k-r}{r+1} \cdot \frac{p}{1-p} = \frac{(k-r)\,b}{(r+1)(n-b)}.$$

Since the denominator is positive, this ratio exceeds 1 exactly when

$$\begin{aligned} & (k-r)b > (r+1)(n-b) \\ \iff \quad & kb - rb > rn - rb + n - b \\ \iff \quad & (k+1)b > (r+1)n \\ \iff \quad & r+1 < \frac{(k+1)b}{n}. \end{aligned}$$

In the same way the ratio equals 1 exactly when $r+1=m$ and is less than 1 exactly when $r+1>m$, where

$$m=\frac{(k+1)b}{n}.$$

So, as r increases, $\mathrm{P}(X=r)$ strictly increases at each step that ends at a value $r+1<m$, and strictly decreases at each step ending at $r+1>m$. Note that $0<m<k+1$ because $0<b<n$, so the switch happens inside the range $0,\ldots,k$.

Ask yourself Where exactly is the largest probability if m is not a whole number?

If m is not an integer, write $M=\lfloor m\rfloor$, so $M<m<M+1$ and $0\le M\le k$. The probabilities strictly increase up to $\mathrm{P}(X=M)$ (every step ends at a value $\le M<m$) and strictly decrease after it (every later step ends at a value $\ge M+1>m$). Hence the unique most probable number of black balls is

$$\left\lfloor\frac{(k+1)b}{n}\right\rfloor,$$

as required.

When it isn't unique. If m is an integer, the step from $r=m-1$ to $r=m$ has ratio exactly 1, so $\mathrm{P}(X=m-1)=\mathrm{P}(X=m)$, and these two equal values are larger than all the others (the probabilities strictly rise to $m-1$ and strictly fall after m). Since $1\le m\le k$, both values are possible. So the answer is not unique precisely when $(k+1)b/n$ is an integer (with $0<b<n$), and then $m-1$ and m are equally most probable. For example, $n=3$, $b=1$, $k=5$ gives $m=2$, and indeed $\mathrm{P}(X=1)=\mathrm{P}(X=2)=\frac{80}{243}$.

The trivial cases sit outside this: if $b=0$ then $X=0$ for certain, which agrees with the formula; if $b=n$ then $X=k$ for certain, and the formula (which would give $k+1$) doesn't apply, because the ratio above involves dividing by $n-b$.

Part (ii)

Without replacement, X has the hypergeometric distribution: choosing the sample as an unordered set of k balls, all $\binom{n}{k}$ sets equally likely,

$$\mathrm{P}(X=r)=\frac{\binom{b}{r}\binom{n-b}{k-r}}{\binom{n}{k}}.$$

Because $k\le b$, the largest possible value is $r=k$; the smallest is $r_0=\max(0,\,k-(n-b))$, since we can't draw more than $n-b$ non-black balls. For $r_0\le r\le k$ the probability is positive, and for $r_0\le r\le k-1$, using $\binom{a}{r+1}=\binom{a}{r}\frac{a-r}{r+1}$ and $\binom{a}{s-1}=\binom{a}{s}\frac{s}{a-s+1}$,

$$\frac{\mathrm{P}(X=r+1)}{\mathrm{P}(X=r)}=\frac{b-r}{r+1}\cdot\frac{k-r}{n-b-k+r+1}.$$

Both denominators are positive in this range, so the ratio exceeds 1 exactly when

$$(b-r)(k-r) > (r+1)(n-b-k+r+1).$$

Expanding, the left side is $bk - r(b+k) + r^2$ and the right side is $(r+1)(n-b-k+1) + r^2 + r$. The r^2 terms cancel, and the inequality becomes

$$\begin{aligned} & bk - r(b+k) > r(n-b-k+1) + (n-b-k+1) + r \\ \iff \quad & bk + b + k + 1 > r(n+2) + (n+2) \\ \iff \quad & r + 1 < \frac{(b+1)(k+1)}{n+2}. \end{aligned}$$

Ask yourself Does the switch point actually lie inside the range of possible values $r_0, \ldots, k$?

Let $m = \dfrac{(b+1)(k+1)}{n+2}$. Since $b+1 < n+2$ we have $m < k+1$. For the lower end, $m > 0$ trivially, and if $r_0 = k+b-n > 0$ we need $(b+1)(k+1) > (n+2)(k+b-n)$. Writing $x = n-b$ and $y = n-k$, the left side minus the right side is

$$(n+1-x)(n+1-y) - [(n+1)^2 - 1 - (n+2)(x+y)] = 1 + x + y + xy > 0.$$

So $r_0 < m < k+1$ in every case, and the argument of part (i) goes through word for word: the probabilities strictly rise while $r+1 < m$ and strictly fall once $r+1 > m$. Hence, when it is unique, the most probable number of black balls is

$$\left\lfloor \frac{(b+1)(k+1)}{n+2} \right\rfloor.$$

It is not unique exactly when $(b+1)(k+1)/(n+2)$ is an integer; then $m-1$ and m (both possible values, since $r_0 \leq m-1$ and $m \leq k$) are equally likely and more likely than anything else. For example, $n = 10$, $b = 7$, $k = 5$ gives $m = 4$, and $X = 3$ and $X = 4$ are tied, each with probability $\frac{105}{252} = \frac{5}{12}$.

Checks

- By computer we computed both distributions exactly with fractions for every $n \leq 24$, $0 \leq b \leq n$ and $1 \leq k \leq 19$ (with $k \leq b$ in part (ii)), found the modes by brute force and confirmed both formulas and both tie conditions in all cases (2565 cases for part (ii)).
- For $n = 10$, $b = 6$, $k = 4$: $m = \frac{35}{12}$, so the formula gives 2, and $X = 2$ is indeed the unique mode.
- If n is huge compared with k, removing a few balls barely changes the proportion of black balls, so part (ii) should look like part (i). It does: $(b+1)(k+1)/(n+2) \approx (k+1)b/n$.

Common slips

- Concluding that the mode is $\lfloor kb/n \rfloor$, the integer part of the mean. The mode is close to the mean but not always equal to its integer part.
- Forgetting that when the ratio is exactly 1 there are two modes, not one.
- In part (ii), slipping a sign when expanding $(r+1)(n-b-k+r+1)$; the cancelling r^2 terms are a useful check.

Beyond STEP

Comparing successive probabilities is the standard way to locate the mode of any discrete distribution, and the same calculation gives the Poisson mode $\lfloor \lambda \rfloor$ as a limiting case of part (i). The hypergeometric distribution is central to sampling theory and to Fisher's exact test in statistics. Distributions whose probabilities rise and then fall like this are called unimodal; for the binomial and hypergeometric this comes from a stronger property, log-concavity. For more, see Grimmett and Welsh, *Probability: An Introduction*, or Blitzstein and Hwang, *Introduction to Probability*.

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